Predicting Fulvestrant Sensitivity in Cancer Cell Lines: A Baseline Model Using GDSC Data

1. OBJECTIVE/QUESTION

   When modeled individually using molecular availability and cancer-type metadata instead of averaging across populations; can the sensitivity(measured as AUC) of the drug Fulvestrant be predicted in GDSC2 cell lines?

2. DATA SOURCE

   GDSC2 dataset (hosted on Kaggle, generated by Wellcome Sanger Institute as a part of the Genomics of Drug Sensitivity in Cancer programme), verified 242035 rows against the official GDSC2 release, filtered by DRUG_NAME to Fulvestrant -1680 rows.

3. DATA CLEANING & PREPROCESSING

   The same 65 rows were missing CNA, Gene Expression and Methylation were dropped (1680 to 1615 rows that retained 96%)
   MSI nulls(19) kept as a category “Unknown”
   Cancer type was selected because drug sensitivity varies enormously by cancer type, checked for null values and obtained 294/1615 (~18%), not dropped but kept in Unknown category,
   Cancer Type had 32 distinct categories obtained, Types with fewer than 20 cell lines were grouped into a single “Other” category ,reducing 32 to 26 .Kept “Unknown” and “Unable To Classify separately”.

4. FEATURE ENGINEERING

Feature matrix (X) selection involved certain considerations:
Anything from the data available that is response derived is not a prediction, therefore AUC, LN_IC50, Z_SCORE are not selected.
Dropped DRUG_ID, DRUG_NAME, TARGET, TARGET_PATHWAY; because Fulvestrant was filtered.
Identifiers were also dropped : COSMIC_ID, CELL_LINE_NAME
Kept: CNA, MSI, Cancer Type : these 5 categories went through decluttering, encoding work.(encoding details covered below.)
Performed pd.get_dummies on the 5 to create binary columns per category, for encoding. They are categorical because they are availability flags and not continuous measurements. All five take a discrete set of labels, rather than a range of numbers; that is why they need one hot encoding.
LIMITATION: All features kept are only proxies, (availability flags, status, a tissue type label), not molecular measurements, so the model's feature space is narrower than genomic features.

5. MODEL

Selected AUC as target variable (y). AUC is a continuous drug sensitivity readout - lower the AUC, higher the drug sensitivity.
Chose RandomForestRegressor, random forest is used because it handles a mix of one hot encoded categorical features.
Train/Test split: 80/20 split was used. For reproducibility, random_state was set to 42.
A mean-predictor baseline (scikit-learn DummyRegressor) was fitted on the same train/test split for comparison.

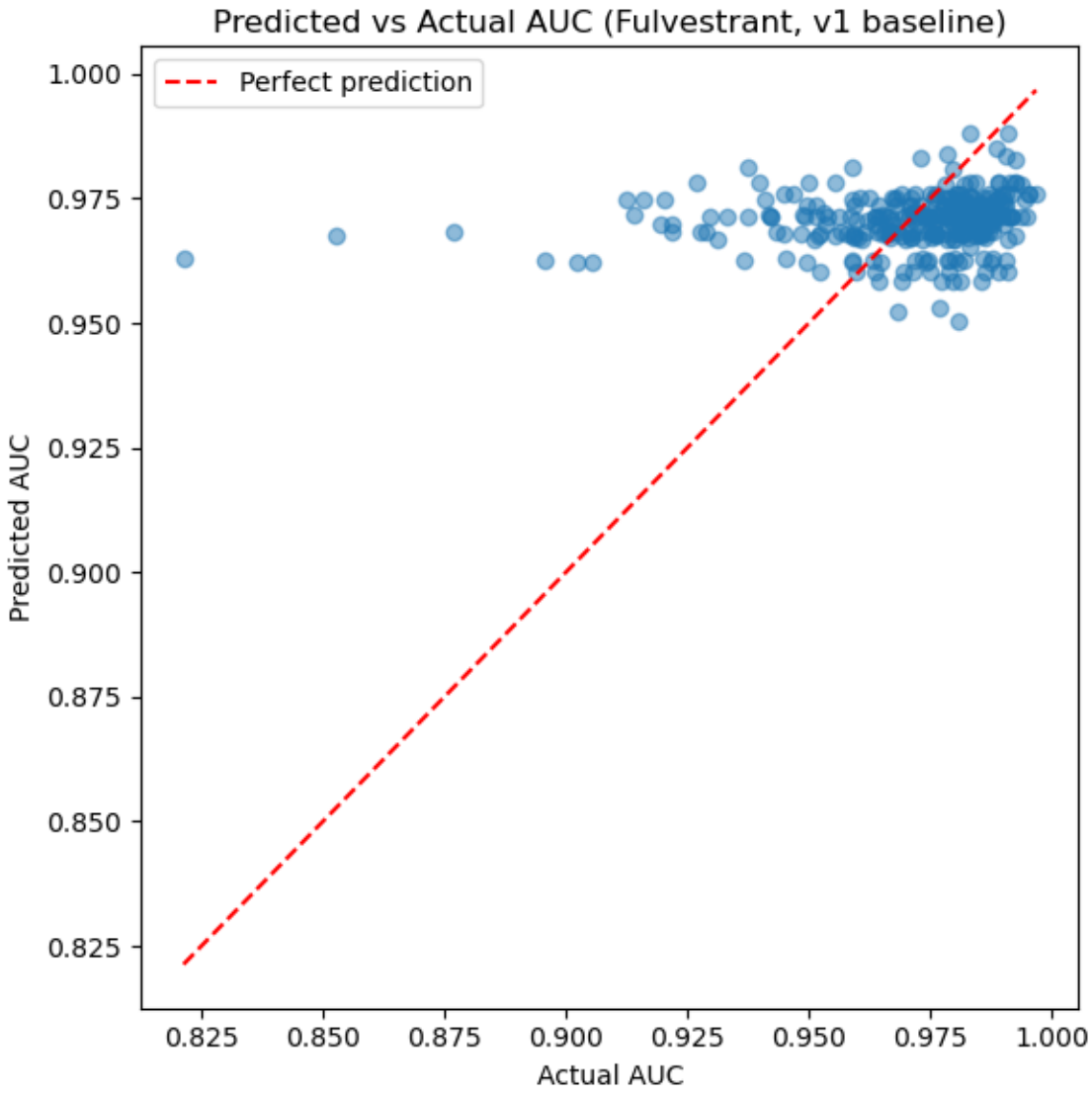
6. RESULTS

The model was evaluated on a held-out test set of 323 cell lines (20% split, random_state = 42).

Mean predictor baseline (DummyRegressor) - R^2 was -0.001 and MSE was 0.000457
RandomForestRegressor - R^2 was 0.017 and MSE was 0.000448

The Random Forest explains about 1.7% of the variance in AUC across cell lines, so about 98.3% of the variation in Fulvestrant sensitivity is not captured by cancer type and availability flags. It lowers MSE by about 2% compared with always predicting the average. On a single test split, this small improvement cannot be distinguished from noise.
The highest-ranked feature was Cancer Type_BRCA (importance 0.145, about 1.8x the next highest, Cancer Type_LIHC at 0.081). BRCA is the fourth largest group (n = 91 of 1,615), and larger groups such as LUAD (n = 108) did not appear in the top 10 features, so group size alone does not explain its rank. Among the five largest cancer-type groups, BRCA had the lowest mean AUC (0.963, against 0.968 to 0.974) and the widest spread (SD 0.029, against about 0.019 to 0.022). Some smaller groups had similar or lower mean AUC (LIHC 0.962, STAD 0.963). This is compatible with Fulvestrant's ER-targeted use in breast cancer, but the differences in mean AUC are small, so it should be treated as a hint rather than a confirmation. Feature importance also shows only which features the model used, not the direction of their effect.
A scatter plot of predicted against actual AUC showed predicted values squeezed into a narrow band (about 0.96 to 0.98), while actual AUC spanned about 0.82 to 1.0. This is

visual evidence that the model cannot distinguish sensitive from resistant cell lines



7. INTERPRETATION / DISCUSSION

The model's weak R^2 (0.017) is best explained by a data-granularity limitation which was discovered during cleaning: CNA, Gene Expression and Methylation were expected to be

molecular measurements, but were found to be binary availability flags (Y/N) rather than continuous values.
This means the model didn't have genuine molecular signals to learn from. The weak performance therefore reflects a limitation in feature resolution, other models and hyperparameter tuning were not tested.

Cancer type + molecular-data-availability flags alone carry almost no signal for predicting drug sensitivity to Fulvestrant — this empirically suggests that real molecular measurements (actual expression levels, mutation calls, methylation values), not just whether that data exists, may be needed for meaningful drug response prediction.

The model's top feature (BRCA) is compatible with Fulvestrant's ER-targeted use, but mean AUC differed little between cancer types, so this is a hint rather than a confirmation. It emerged even without finer molecular resolution.
This is consistent in direction with Qureshi et al.(Paper 3), where fine grained MD-simulations derived molecular features substantially outperformed clinical/ demographic data in predicting drug response.

8. LIMITATIONS

This model has several limitations that constrain how its results should be interpreted.

1. Feature granularity: CNA, Gene Expression and Methylation were expected to be real molecular measurements but found to be binary availability flags, not continuous values. The model was never given genuine molecular signal to learn from- only proxies for whether that data exists.
2. Small feature set: Only five features(CNA, Gene Expression, Methylation, MSI, Cancer Type) were available for modeling. No fine-grained genomic data such as copy number values,

expression levels or methylation percentages was accessible in this dataset.

3. Single-drug scope: The model was built and evaluated only for Fulvestrant. It has not been validated on other drugs; Oxaliplatin was identified as a candidate for extending this pipeline but was not run in this version.
4. Weak explanatory power: R^2 (0.017) means the model explains very little of the variance in AUC, and it is only marginally better than a mean-predictor baseline (MSE 0.000448 against 0.000457). As such, this model is not suitable for real predictive use in its current form. Its value here is as a proof-of-concept and diagnostic exercise, not a deployable tool.
5. Missing-data proxy category: Cancer Type had ~18% missing values(294/1615), kept as an “Unknown” category rather than dropped or imputed. Since Cancer Type turned out to be the model's most biologically meaningful feature, this placeholder label in a significant share of rows could dilute the very signal (Cancer Type_BRCA) that drove the model's strongest predictions.

9. FUTURE WORK

Several extensions were identified but not pursued in this v1 model. Oxaliplatin, the second drug scoped alongside Fulvestrant from the start of this project, was not yet run and remains a natural next step for extending this same pipeline. Incorporating real GDSC omics matrices — actual molecular measurement values rather than the Y/N availability flags used here — would directly address the feature-granularity limitation identified above, if time permits. CCLE was considered as an additional dataset and was never formally ruled out; it could be

used to extend or validate these findings further. Finally, a Linear Regression comparison against the RandomForest model remains an open, undecided option as a baseline check.